

CIS 106 – Loops Part 2

For each problem prepare an IPO chart. Then write the code for each. Save the IPO within this document and upload to your repository. After code is complete upload the files (.py) to your repository. Paste the link to your repository into the assignment completion link in Blackboard.

1. Allow the user to enter a principle amount and interest rate repeatedly (need a loop to control the program execution). Compute the annual interest (principle x rate). Compute ending balance to be principle (beginning balance + interest). Display year, beginning balance and ending balance for each of the 5 years. Display the accumulated interest for the 5 years. Note: the new balance by year (this will be the principle for the following year. Format the output.

input	processing	Output
Principal amount	Set year = 1	Year
Interest rate	Loop for 5 years	Beginning balance
User choice to continue	Interest = balance x rate	Ending balance
	Ending = beginning + interest	Total interest
	Accumulate interest	
	Repeat until user stops	

Example:

Enter principle amount: 10000.00

Enter interest rate: 0.10

Year	Beginning Balance	Ending Balance
1	\$10,000.00	\$11,000.00
2	\$11,000.00	\$12,100.00
3	\$12,100.00	\$13,310.00
4	\$13,310.00	\$14,641.00
5	\$14,641.00	\$16,105.00

Total interest earned: \$6,156.00

2. Fibonacci sequence is a sequence of natural order. The sequence is:
1, 1, 2, 3, 5, 8 etc
Use of for loop compute and display first 20 numbers in the sequence. Hint: start with 1, 1.

input	processing	output
none	Initialize first = 1	First 20 numbers
	Initialize second = 1	Fibonacci sequence
	Loop 20 times	
	Next = first + second	
	Update values	

3. Create a text file that contains employee last name and salary. Read in this data. Determine the bonus rate based on the chart below. Use that rate to compute bonus. For each line display the employee last name, salary and bonus. After the loop display the sum of all bonuses paid out.

Input	processing	output
Employee name	Read file line by line	
salary	Determine bonus rate	Name
	Complete bonus	Salary
	Accumulate total bonuses	Bonus
	Repeat for all employees	Total bonuses

Salary	Bonus Rate
100,000.00 and up	20%
50,000.00	15%
All other salaries	10%

Example file (create your own data with at least 5 lines:

Adams
50000.00
Baker
75000.00
Smith
45000.00
Etc

4. Create a text file with item, quantity and price. Read through the file one line at a time. Compute the extended price (quantity x price). For each line display the item, quantity, price

and extended price. After the loop display the sum of all the extended prices, the count of the number of orders and the average order.

input	processing	output
	Read file line by line	Item
Item	Extended = qty x price	Quantity
Quantity	Accumulate total	Price
price	Count orders	Extended price
	Compute average	Total, count, average

Example Data File

Widget

10

50

Hammer

2

10

Saw

4

8

Etc

5. Create a text file with student last name, district code (I or O) and number of credits taken. Compute tuition owed (credits taken x cost per credit). Cost per credit for in district students (district code I) is 250.00. Out of district students pay 500.00 per credit. For each line display student last name, credits taken and tuition owed. After the loop display sum of all tuition owed and the number of students.

input	processing	output
Student name	Read file	Name
District code	Determine cost per credit	Credits
credits	Tuition = credits x cost	Tuition
	Accumulate tuition	Total tuition
	Count students	Student count

Example file

Jones

I

12

Adams

I

10

Baker

O

12

Smith

O

16